# MAZEN YOUNIS

Software Developer | Embedded Systems | Industry 4.0 | AI & RAG Systems

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GitHub | LinkedIn
Portfolio

## SUMMARY

Software Developer with hands-on experience in Industry 4.0, embedded and distributed systems, industrial communication and AI-based applications. Currently pursuing an M.Eng. in Computer Engineering – Embedded Systems, with hands-on experience in Python, OPC UA, system integration, Large Language Models and Retrieval-Augmented Generation (RAG). Focused on developing reliable software and intelligent systems for real-world industrial applications.

## EDUCATION

### M.Eng. Computer Engineering – Embedded Systems

Berliner Hochschule für Technik | Mar 2026 – Sep 2027 (expected)
Embedded and distributed systems, real-time systems, AI and intelligent systems

### B.Eng. Computer Engineering – Embedded Systems

Berliner Hochschule für Technik | Oct 2022 – Mar 2026
Embedded systems, software development, hardware-software integration and real-time systems

### B.Eng. Civil & Environmental Engineering

Beirut Arab University | Sep 2013 – May 2018
Structural and environmental engineering, technical planning and project-based engineering

## TECHNICAL SKILLS

**Programming:** Python, C++, MATLAB

**Embedded & Industrial:** Embedded Systems, OPC UA, Asset Administration Shell (AAS), Distributed Systems, Real-Time Systems

**AI & LLMs:** Large Language Models, Retrieval-Augmented Generation (RAG), Fine-Tuning, LoRA, Alignment & Evaluation

**Software & Tools:** Git, CMake, Linux, VS Code

**Engineering:** System Integration, Industrial Communication, Industry 4.0

## SELECTED PROJECTS

**Master's Thesis – Reliable RAG & LLM Systems**
Fine-tuning, LoRA, alignment, hallucination reduction and evaluation

**Bachelor's Thesis – Industrial Digital Twin Integration**
OPC UA, Asset Administration Shell (AAS) and system integration

## PROFESSIONAL EXPERIENCE

### Software Developer Working Student

Fraunhofer IPK | Jun 2025 – Present | Berlin

- Develop Industry 4.0 software using Python and OPC UA across embedded and distributed systems, system integration and industrial communication.
- Apply Large Language Models and Retrieval-Augmented Generation (RAG), including methods for improving AI reliability and evaluation.
- Contribute to research-oriented software development, testing, debugging and technical documentation in interdisciplinary environments.

### Industry 4.0 Software Development Intern

Fraunhofer IPK | Jan 2025 – Jun 2025 | Berlin

- Contributed to software development and research activities in Industry 4.0, artificial intelligence and industrial digitalisation.
- Supported the implementation, integration and testing of software solutions for industrial applications.

### Sales Representative

Messe Berlin | Jan 2024 – May 2025 | Berlin

- Supported customer communication and sales activities in a fast-paced international event environment.

### Project Manager & Site Manager

M.O.R.A for Engineering & Construction | Sep 2017 – Jan 2020 | Beirut

- Managed engineering projects from planning through execution, coordinating technical requirements, site activities and project stakeholders.
- Led client communication, on-site problem-solving and project coordination while ensuring quality, safety and environmental requirements.

## LANGUAGES

Arabic: Native
English: C1 (IELTS)
German: C1 (DSH)
French: Basic